

AMIT DAS

Frontend Developer | React, Next.js, TypeScript, Node.js

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PROFESSIONAL SUMMARY

Frontend Developer with 4+ years of experience building real-time, data-intensive web and mobile applications across environmental monitoring, civic infrastructure and enterprise procurement. Specialized in React, Next.js and TypeScript, with production depth in WebSocket data streaming, virtualized data grids, workflow-driven interfaces and frontend performance optimization. Currently expanding into full-stack ownership with Node.js and Express. Seeking frontend or full-stack developer roles centered on real-time, data-heavy products.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Java, HTML5, CSS3, SQL

Frontend: React.js, Next.js, Remix, Redux Toolkit, React Query (TanStack Query), Zustand, Tailwind CSS, shadcn/ui, Vite

Mobile: React Native, Expo, NFC Integration, Offline-First Sync, Geolocation

Backend & APIs: Node.js, Express.js, REST APIs, WebSockets (Socket.IO), JWT Authentication

Data & Visualization: Real-Time Data Streaming, Interactive Charting, Interactive Mapping, Virtualized Data Grids, Role-Based Access Control, CSV/Excel Export Pipelines

Performance: Code Splitting, Lazy Loading, Memoization, List Virtualization, Caching and Request Deduplication, SSR/ISR, Core Web Vitals

Tools & Practices: Docker, Git, GitHub, Linux, Nginx, Vercel, CI/CD, Jira, Agile/Scrum, Component-Driven Development, Design Systems, Code Review

PROFESSIONAL EXPERIENCE

Software Developer (Frontend)

Distronix | Salt Lake, Kolkata, India | Jul 2022 – Present

Own the UI layer on data-heavy platforms where live hardware telemetry and enterprise workflows meet the browser: architecture, performance, component systems and delivery across web and mobile.

- Migrated legacy dashboards to a React and TypeScript architecture, reducing initial load time from approximately 6 seconds to under 2 seconds and eliminating full-page refreshes.
- Designed and shipped a reusable library of 30+ themeable components adopted across the product suite, reducing new-screen build time from days to hours and standardizing UI patterns between teams.
- Applied code splitting, route-level lazy loading and memoization across the core product, raising Lighthouse performance scores from the 50s to 90+.
- Built REST API endpoints with Node.js and Express and established shared TypeScript types across services, reducing runtime integration defects and improving API contract reliability.
- Containerized the frontend stack with Docker, reducing new-developer environment setup from half a day to a single command.
- Delivered SEO-optimized marketing and product sites in Next.js and Remix using SSR and ISR, improving Core Web Vitals and organic search visibility.
- Defined frontend conventions for project structure and state management, led code reviews, and collaborated with backend, QA and product teams in an Agile workflow.

SELECTED PROJECTS

CAAQMS — Continuous Ambient Air Quality Monitoring System

Distronix | 2025 – 2026 | Next.js, TypeScript, React, Redux, React Query, Tailwind CSS, REST APIs

Regulatory-grade environmental monitoring platform for real-time and historical air quality surveillance across 25+ monitoring stations.

- Built a data-dense monitoring dashboard tracking 12 pollutant and meteorological parameters (PM2.5, PM10, SO2, NO2, CO, O3, NH3, Benzene) on a 15-minute station reporting cadence.

- Developed an interactive charting suite (line, bar, scatter) with drill-down from station overview to individual parameter trends, enabling analysts to move from anomaly to root cause in a single view.
- Engineered virtualized data grids rendering 10,000+ rows at a sustained 60fps using windowing and memoization, replacing tables that stalled beyond approximately 1,500 records.
- Built a historical data module with multi-parameter filtering and date-range selection across a rolling archive of 10M+ records, plus a CSV and Excel export pipeline formatted for statutory pollution control board submissions.
- Implemented React Query caching, background refetching and request deduplication for an all-day dashboard, cutting redundant network calls by approximately 60%, and added an analyzer diagnostic module surfacing instrument health and calibration status.
- **Impact:** replaced end-of-cycle reporting with live cross-station visibility and made year-scale datasets explorable in the browser rather than exported and analyzed offline.

Vehicle Tracking System — Municipal Solid Waste Management

Distronix | 2023 – 2025 | React, Vite, Tailwind CSS, Socket.IO, React Native, Expo, NFC

Real-time fleet tracking and field-verification platform for municipal solid waste operations, covering 20+ collection vehicles across 12 wards.

- Built a live operations dashboard streaming vehicle location, speed and status at 2–3 second intervals over Socket.IO, handling 50+ concurrent device streams with automatic reconnection.
- Developed an interactive map with real-time positioning, route-history playback and status-based color coding (Active, Idle, Offline) for at-a-glance fleet health.
- Implemented an NFC tap-to-verify collection flow processing approximately 500 scans per day, replacing paper logs with timestamped, location-bound proof of service.
- Shipped a React Native and Expo application used by 45+ field workers, featuring geotagged photo capture and offline-first sync that reconciles queued scans on reconnect, optimized for the low-end 2 GB RAM Android devices standard in field deployment.
- **Impact:** reduced missed-pickup detection from end-of-day reporting to under 5 minutes and eliminated paper-based tracking along with its transcription errors.

Inventory and Procurement Management System

Distronix | 2022 – 2024 | React, Vite, TypeScript, Tailwind CSS, REST APIs

Enterprise inventory and procurement platform digitizing the full purchase-to-stock lifecycle for internal stores, finance and operations teams.

- Built end-to-end purchase order workflows covering PO creation, multi-level approval, amendment and revision tracking, and shipment receipt reconciled against ordered quantities.
- Developed requisition and indent modules allowing departments to raise material requests routed through configurable approval chains before conversion into purchase orders.
- Implemented stock management and stock disposal modules with live inventory levels, goods-receipt reconciliation (ordered, received and pending), and audit-tracked write-offs.
- Designed role-based interfaces separating requester, approver, store-keeper and administrator permissions, ensuring each user saw only the actions valid at their stage of the workflow.
- Built dense, filterable data tables with server-side pagination, search and status filtering for browsing purchase orders, stock ledgers and transaction history.
- **Impact:** replaced spreadsheet and paper-based procurement tracking with a single auditable system, giving stores and finance a shared, real-time view of order status from requisition through receipt.

EDUCATION

Bachelor of Technology (B.Tech), Information Technology

University of Calcutta | Aug 2018 – Jun 2022 | CGPA: 8.05 / 10

Higher Secondary Certificate, Science

Jhargram Banitirtha High School | 2015 – 2017 | 86.4%